

Manual Brush vs Artificial Intelligence Pencil: Evaluating tools for auto-contour refinement of head-and-neck tumors on Positron Emission Tomography and Computed Tomography scans

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Abstract

Background and Purpose: To resolve errors in auto-contours, clinicians currently use a manual brush tool. These are inefficient, especially for larger errors, where each incorrect pixel has to be rectified. An alternative is Artificial Intelligence (AI)-assisted contour refinement using visual cues such as pencil strokes (or scribbles) drawn within inaccurate regions. However, no comparison against the widely used manual brush exists, especially in a study with clinical users. Thus, the goal of this study was to compare the time-efficiency and contour quality of the manual brush against the AI pencil.

Materials and Methods: A web-based interface and an AI pencil trained on 452 scans from the HECKTOR2022 dataset was designed to conduct auto-contour refinement sessions. Both tumor contouring experts (N= 4) and non-experts (N= 7) performed contour refinement with data from 6 patients. The AI pencil supports two-dimensional (2D) interactions (i.e., scribbles) to refine three-dimensional (3D) tumor contours on head-and-neck Computed Tomography + Positron Emission Tomography scans. The efficiency (time) and effectiveness (Dice / surface Dice @ 2 mm) of the manual brush and AI pencil were compared.

Results: For tumor auto-contour refinement, the AI pencil was [5% – 78%] (median=258.5 s) faster across 42 non-expert sessions and [16% – 97%] (median=234 s) faster across 24 expert sessions. The average inter-observer variability (calculated by Dice / surface Dice @ 2 mm) across 6 patients was equivalent between the manual brush (0.89/0.90) and AI pencil (0.90/0.92) for the expert sessions.

Conclusions: The AI pencil offers a promising alternative to traditional manual brushes in auto-contouring-based radiotherapy workflows. It improved the time efficiency while maintaining final contour quality for auto-contour refinement.

1. Introduction

Auto-contouring in radiotherapy has made great progress over the last 10 years with improvements in deep-learning based Artificial Intelligence (AI) models and the proliferation of clinically-available commercial tools [1–5]. Widespread use of these AI-based auto-contouring tools can be attributed to the time gains they provide. However, as these tools are still imperfect, clinicians currently perform a time-consuming manual quality assessment (QA) and refinement step. This bottleneck offsets some of the time gains provided by auto-contouring.

A limited number of automated techniques have been proposed to reduce the auto-contour refinement bottleneck by either error-detection [6–8] or error-correction [9–13]. This work focused on error-correction wherein a user provided sparse feedback iteratively to improve an imperfect auto-contour. This feedback is usually in the form of sparse visual cues such as pencil strokes (in the form of dots or scribbles) in the erroneous regions to rectify them. Dots and scribbles can be considered sparse interactions when compared to the more dense brush interaction where all the erroneous voxels need to be rectified. Literature on contouring with sparse user input has mostly reported on single-step auto-contouring with two-dimensional (2D) [14–16] or three-dimensional (3D) [17–22] models with only a few studies reporting on results of iterative contour refinement [9–13]. Two of the iterative contour refinement studies conducted a study with clinical users and reported time savings [11, 12] with models that take a single modality as input. Since time-based evaluation with real clinical users is important, this work built a lightweight and multi-modal 3D model and also tracked the evolution of contouring metrics as more interactions were provided by the user.

Thus, the main aim of this study was to compare the time efficiency and contour quality of auto-contour refinement with human users across two refinement tools i.e., manual brush and AI pencil. A further aim was to assess how users’ interactions and refinement outcomes varied between the two tools throughout the iterative refinement process.

2. Materials and methods

To compare the two contour refinement tools, a head-and-neck tumor dataset was used to train an auto-contouring and contour-refinement model (i.e., AI pencil). The contour refinement tools were then evaluated on a web interface by users.

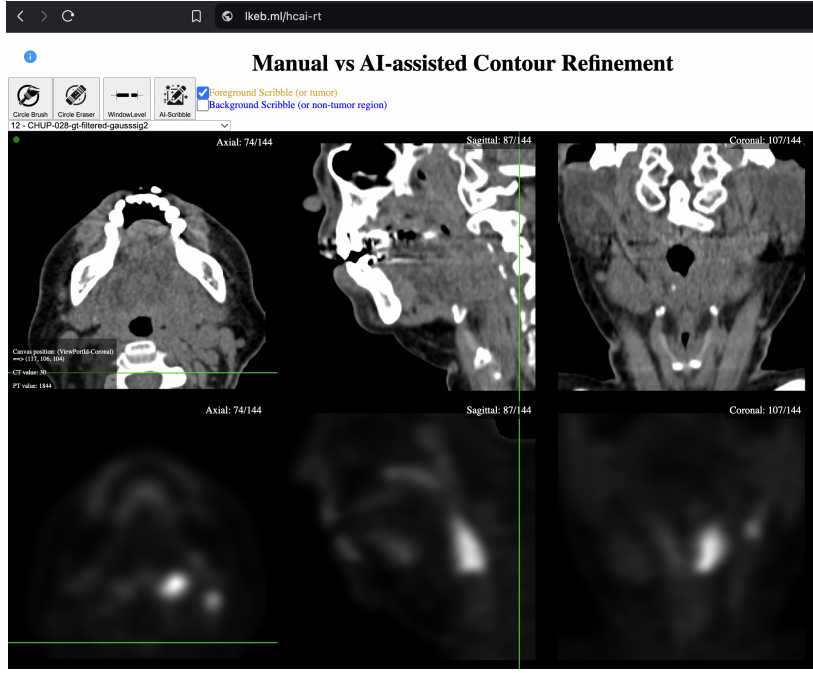
2.1. Dataset

A head-and-neck tumor dataset from the HECKTOR2022 challenge was used [23] which contains 524 pairs of Computed Tomography (CT) and Positron Emission Tomography (PET) scans from seven clinics. The data originated from four countries; data from three of them (Canada, Switzerland, United States of America; 452 pairs) was used for training and validation, and from the remaining country (France; 72 pairs) for testing. More details can be found in [Supplementary Material A](#).

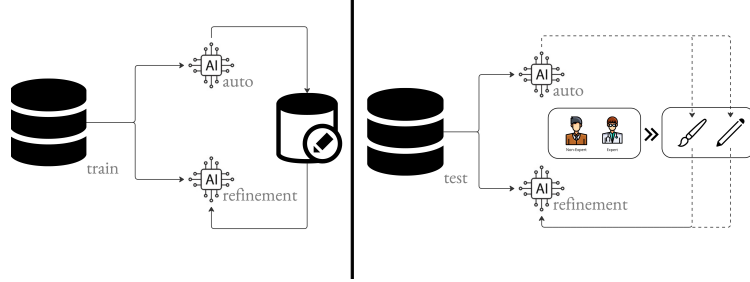
2.2. Auto-contour and contour-refinement model training

A standard UNet architecture ($\sim 1.2\text{M}$ parameters) from the MONAI (Medical Open Network for AI) software package [24] was chosen for both the auto-contouring and the contour-refinement model. Each model was trained using the standard cross-entropy loss. The goal here was not to achieve the best contouring performance, but rather to provide an initial segmentation for contour refinement. More details can be found in [Supplementary Material B](#).

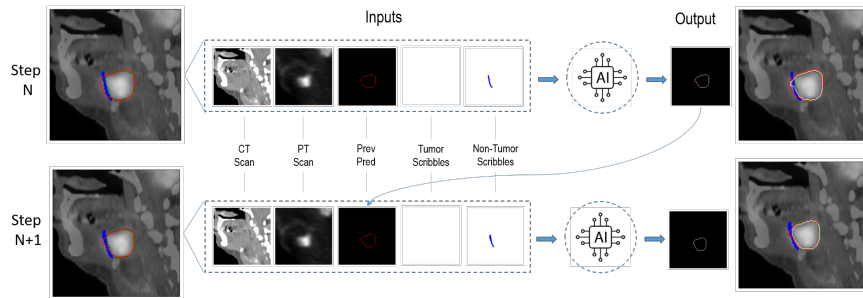
The auto-contouring model took as input the CT and PET scans and was trained, using ground truth annotations, to produce a tumor mask. The contour-refinement model took five inputs: CT, PET, the previously predicted contour, tumor scribbles and non-tumor scribbles and produced a refined contour as output ([Fig. 1c](#)). The tumor scribble was a stroke made on tumor voxels while the non-tumor scribble was made on background voxels. This model was trained using the mask predictions of the auto-contouring model as input. During training human scribbles were simulated by generating logic-based 2D scribbles in the false positive (FP) and false negative regions (FN) of the auto-contour models’ predictions [9]. Depending on the



(a) Web interface for contouring



(b) Training & testing workflow



(c) Neural network design

Fig 1: a) Web interface to perform contour refinement which shows axial/sagittal/coronal views for both PET + CT scans. On the top of the interface, the user can select contour editing tools (manual brush or AI pencil) and a patient from a list. b) Training workflow (left) showing the same database used to train the auto-contouring and contour-refinement models. Testing workflow (right) showing how auto-contours were modified by (non) experts using brushes (manual) or scribbles (AI-assistance). c) Inputs used within the neural network to make contour refinements with ground-truth (green), prediction (red) and refinement (pink).

region (i.e. FP or FN), the scribble was passed either as a tumor (for FN) or non-tumor (for FP) scribble. The model was then trained by comparing the refined contour predictions against the ground truth from the HECKTOR2022 dataset. Note that the word scribble here is used to encompass any other human user inputs such as dots, circles, crosses or other arbitrary lines.

2.3. Auto-contour and contour-refinement model validation

The outputs from both models were evaluated using the standard Dice and surface Dice (@2 mm) metric. The 2 mm threshold was motivated by the Hausdorff Distance 50% (HD50) results in the previous study by Bollen et al [25]. The single-step auto-contouring model produced only one value for these metrics per patient. However, the contour refinement tools - the manual brush and AI pencil (i.e., contour-refinement model) were applied iteratively and hence evaluated at each “interaction” with the above metrics. Here, an interaction referred to a single editing action with a contour tool, beginning when a correction was initiated with a computer mouse and ending when that action was completed. This ensured capture of one complete mouse press-and-release event. A correction was defined as pixels drawn with a manual brush or a stroke made by the AI pencil. In other words, this corresponded to the manual brush doing direct contour editing, whereas for the AI pencil it meant making either a tumor or non-tumor scribble.

2.4. Manual Brush and AI Pencil Comparison

Absolute time was used as the primary measure of tool comparison. Interaction count, pixels drawn, and slices scrolled were logged as secondary descriptors of user workload. Savings provided by the AI pencil when compared to the manual brush were also shown in percentages as:

$$\Delta M(\%) = (M_{\text{manual}} - M_{\text{AI}}) / M_{\text{manual}}, \quad (1)$$

where M was either total time, total interactions, total pixels drawn, or total slices scrolled.

To verify whether both tools produced similar contours, the inter-tool variability (ItV) per patient was computed by comparing the final contours from the manual brush and AI pencil. Moreover, the inter-observer-variability (IoV) [26] was computed for each tool to determine if automation tools lead to standardization. The IoV computed metrics between the contours of multiple observers (using the same tool) and reported the median. A higher value indicated more agreement between the observers. Finally, as is the standard, the final contours were also compared against the dataset ground truth.

2.5. Web-based tool

A web-based user interface (Fig. 1a) was developed using off-the-shelf libraries for the front end (cornerstone3D [27]), back end (FastAPI [28]) and scan database (Orthanc [29]). This interface (video [30], code [31]) provided the manual brush, AI pencil as well as panning, zooming and scrolling capabilities for both the registered PET and CT scans. Shortcuts were provided to show/unshow contours, to change the size of the manual brush, as well as to switch between the foreground (tumor) and background (non-tumor) scribbles of the AI pencil. For more details refer to [Supplementary Material C](#).

2.6. User Cohort

To compare the manual brush against the AI pencil, four head-and-neck tumor contouring experts (radiation oncologists with 2/4/11/21 years of experience) and seven non-experts (PhD candidates in AI for medical imaging) participated in this study. Contour refinement sessions were conducted with non-experts to first establish a proof of principle for the model behind the AI pencil. To support the non-experts, the ground truth tumor contour was given to them as a reference, and they were tasked to refine towards it. The experts were not shown the reference contours, and they were tasked with refining based on their expert opinion.

In each session, the user was assigned a patient and a contour-refinement tool (manual brush or AI pencil, see Fig. 1b). All users initially underwent training sessions with four patients from the validation dataset before the start of the study. Similar to previous studies [11], there was a gap of at least one week

between the manual brush and AI pencil sessions of a user-patient pair. This reduced potential learning effects on the anatomy of that patient.

To compute the evaluation metrics, each interaction of the non-experts was compared against the reference contour, while each interaction of the experts was compared against their final contour. Consequently, the final metric value for the experts would always be 1.0.

Table 1: Time (a,b) and interaction count (c,d) savings provided by the AI pencil when compared to manual brush for experts (E) and non-experts (NE) across patients (P). Time savings were in seconds (s) and the percentages (%) indicated how fast the AI pencil was when compared to the manual brush (i.e., $(T_{\text{Manual}} - T_{\text{AI}})/T_{\text{Manual}}$).

(a) Time savings (seconds, percentage) provided by the AI pencil in non-expert (NE) sessions

	P1	P2	P3	P4	P5	P6
NE1	536 (78%)	1018 (65%)	322 (29%)	328 (46%)	96 (20%)	722 (52%)
NE2	326 (68%)	403 (57%)	440 (64%)	167 (38%)	112 (31%)	514 (56%)
NE3	319 (57%)	402 (55%)	391 (57%)	197 (47%)	223 (36%)	676 (58%)
NE4	90 (22%)	125 (24%)	194 (40%)	180 (54%)	130 (49%)	110 (19%)
NE5	112 (25%)	324 (50%)	96 (29%)	271 (54%)	201 (61%)	52 (10%)
NE6	440 (56%)	494 (78%)	229 (51%)	189 (38%)	340 (70%)	826 (61%)
NE7	199 (33%)	121 (21%)	289 (41%)	17 (5%)	246 (49%)	305 (43%)
Range	[90,536]	[121,1018]	[96,440]	[17,328]	[96,340]	[52,826]
(min, max)	([22%,78%])	([21%,78%])	([29%,64%])	([5%,54%])	([20%,70%])	([10%,61%])

(b) Time savings (seconds, percentage) provided by the AI pencil in expert (E) sessions.

	P1	P2	P3	P4	P5	P6
E1	276 (54%)	135 (52%)	52 (20%)	676 (86%)	408 (88%)	775 (97%)
E2	504 (70%)	307 (70%)	138 (42%)	304 (63%)	361 (85%)	666 (76%)
E3	230 (50%)	151 (73%)	29 (16%)	251 (85%)	238 (75%)	223 (68%)
E4	118 (40%)	71 (47%)	46 (34%)	222 (78%)	55 (83%)	292 (78%)
Range	[118,504]	[71,307]	[29,138]	[222,676]	[55,408]	[223,775]
(min, max)	([40%,70%])	([47%,73%])	([16%,42%])	([63%,86%])	([75%,88%])	([68%,97%])

(c) Interaction savings (count, percentage) provided by the AI pencil in non-expert (NE) sessions.

	P1	P2	P3	P4	P5	P6
NE1	97 (59%)	103 (61%)	136 (69%)	88 (67%)	49 (56%)	202 (73%)
NE2	56 (58%)	60 (66%)	78 (75%)	69 (72%)	43 (67%)	128 (81%)
NE3	105 (74%)	26 (30%)	102 (69%)	73 (70%)	73 (82%)	116 (65%)
NE4	98 (65%)	46 (42%)	170 (80%)	88 (83%)	95 (84%)	178 (78%)
NE5	55 (56%)	8 (14%)	62 (65%)	67 (67%)	64 (81%)	62 (52%)
NE6	130 (73%)	19 (19%)	111 (70%)	65 (65%)	118 (87%)	121 (66%)
NE7	152 (64%)	49 (44%)	109 (71%)	47 (55%)	87 (76%)	158 (75%)
Range	[55,130]	[8,103]	[62,170]	[47,88]	[43,118]	[62,202]
(min, max)	([56%,74%])	([14%,66%])	([65%,80%])	([55%,83%])	([56%,87%])	([52%,81%])

(d) Interaction savings (count, percentage) provided by the AI pencil in expert (E) sessions.

	P1	P2	P3	P4	P5	P6
E1	99 (77%)	19 (55%)	41 (67%)	117 (88%)	117 (95%)	190 (97%)
E2	119 (85%)	40 (78%)	66 (80%)	76 (80%)	69 (86%)	143 (88%)
E3	91 (81%)	18 (72%)	33 (66%)	78 (90%)	57 (90%)	87 (93%)
E4	72 (74%)	38 (84%)	43 (62%)	79 (90%)	26 (92%)	104 (95%)
Range	[72,119]	[19,40]	[33,66]	[76,117]	[26,117]	[87,190]
(min, max)	([74%,85%])	([55%,84%])	([62%,80%])	([80%,90%])	([86%,95%])	([88%,97%])

3. Results

The auto-contouring model achieved an average Dice score of 0.76 and average surface Dice score (@2 mm) of 0.72 on the test set. From this set, five patients were selected with a surface Dice in the [0.65, 0.7] range as cases in need of QA (P1, P2, P4-P6), and one patient with a surface Dice of 0.88 as a high-quality case (P3).

Upon comparing the tools for the refinement of auto-contours, the AI pencil was [5% – 78%] (median = 49% or 258.5 s) faster for non-expert sessions (Table 1a) and [16% – 97%] (median=70% or 234 s) for expert sessions (Table 1b). The same can be seen in line plots of Fig. 2 depicting Dice (surface Dice @ 2 mm) performance across time. The average inference time for the model behind the AI pencil was 2.9 s which was counted in the total time taken by the AI pencil. Also, the AI pencil required [14% – 87%] and [55% – 97%] fewer interactions for non-expert (Table 1c) and expert (Table 1d) sessions, respectively. A retrospective power analysis performed on the paired time-difference data (i.e., 66 sessions combining both expert and non-expert times) indicated a large effect size (Cohen’s $d = 1.39$) and statistical power > 0.99 for $\alpha = 0.05$. This confirmed that the study was sufficiently powered to detect the observed differences.

Additionally, Fig. 3 shows a histogram plot of pixels drawn (for each interaction) during contour refinement which were [63% – 93%] and [81% – 99%] less for the AI pencil than the manual brush for non-expert and expert sessions, respectively (Supplementary Material D). Across the 10 refinement examples in Fig. 4, scribbles made with the AI pencil were used to rectify both false positives (Fig. 4a, 4c, 4d, 4f, 4g, 4h, 4i) and false negatives (Fig. 4b, 4e, 4j). While most scribbles were shorter in length, others were lengthier and explicit with their feedback (Fig. 4f, 4i). Regardless of the style of the scribble, a sparse scribble in 2D (on slice s) propagated its changes in 3D. This was seen when one observed the updated contour on slices $s-1$ and $s+1$. While some interactions aligned the refined contour with the reference contour with a single scribble (Fig. 4g, 4h), others still required more interactions (Fig. 4i, 4j).

The following results for experts are all reported as Dice (surface Dice @ 2 mm). For inter-tool variability (ItV), values in the range of [0.67, 0.89]([0.58, 0.85])(*average* = 0.81(0.77)) (Table 2a) were observed across all patients. The average inter-observer variability (IoV) was 0.89 (0.90) for the manual brush and 0.90 (0.92) for the AI pencil (Table 2b). Finally, comparing with the HECKTOR2022 dataset ground truth led to an average of 0.84 (0.81) for the manual brush and 0.84 (0.85) for the AI pencil. The ground truth contours were reviewed and corresponding values for the original, unreviewed ground truth can be found in Supplementary Material F.

4. Discussion

The widespread adoption of auto-contouring has reduced the contouring bottleneck in radiotherapy. Speeding up QA of these auto-contours will further diminish this bottleneck. This work investigated the use of an AI pencil for this purpose and compared its time-effectiveness, user load and capability for contour standardization against the standard manual brush. The AI pencil could understand sparse visual cues and was able to propagate the 2D cue to 3D updates (Fig. 4). This reduced the total time to QA the auto-contours by a median of 49% (258.5 s) for the non-expert cohort and 70% (234 s) for the expert cohort (Table 1, Fig. 2).

As a proof of principle, auto-contour refinement sessions were first conducted with non-experts. They were shown the reference contour since they needed to be provided with a target contour to work towards (refer to the video in [30]). Since both the manual brush and AI pencil sessions aimed to refine the predicted auto-contour to the same reference, their timing curves can be directly compared. The results (Table 1a, 1c and Fig. 2a - 2f) show that the use of the AI pencil sped up auto-contour refinement with early and obvious advantage in the majority of cases (Fig. 2a, 2b, 2d, 2e, 2f). In this work, time was chosen as a primary metric, while interaction count was considered as a secondary metric. Since the total time taken was partly dependent on the interaction count, this was also reported as a secondary descriptor. However, since the duration and effect of a single interaction can vary between tools, absolute time savings was used as the primary metric. For case Fig. 2c, which was the case with the high initial Dice, an early drop in performance of the AI pencil was observed, recovering from this after 20% of the manual brush interactions. A potential

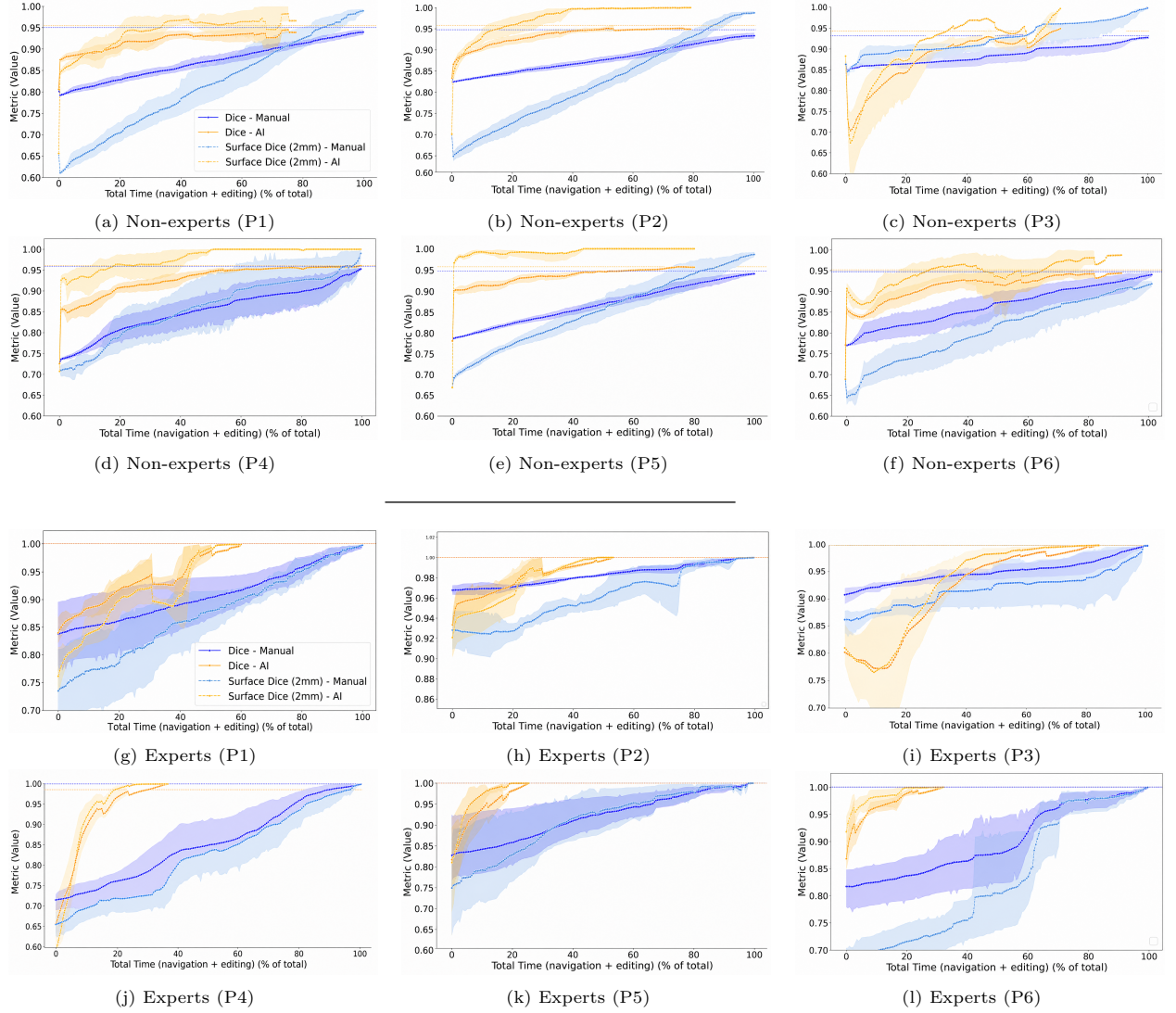


Fig 2: Line plots (with 95% CI) comparing contour refinement sessions for manual brush (in blue) and AI pencil (in orange) tools across 7 non-experts (a-f) and 4 experts (g-l). Here each session corresponds to a user (non-expert/expert) refining one patient (P) using a specific tool. The timing of each session is normalized into the $[0,100]$ range by taking the max time across manual brush and AI pencil sessions and assigning it a value of 100. The normalized time on the x-axis is a combination of the slice navigation and contour editing time for each session.

explanation of this behavior is the fact that the two AI models were not internally aligned, meaning that the first iteration of the AI pencil, i.e. still having little manual input, may default to its own prediction of the contour.

Having established a proof-of-principle with the non-experts, the AI pencil was then tested in a real-world setting where the experts did not see the reference contour. Note that this expert cohort represented the clinically relevant evaluation. In half of the cases (Fig. 2j, 2k, 2l), a clear and early improvement was seen due to the use of the AI pencil. In other cases (Fig. 2g, 2h) smaller or later improvements over the manual brush were seen. And finally, similar to the non-experts, a drop and eventual rise for case Fig. 2i was seen. Note that for these specific figures of experts, their final contour served as the reference standard for each of their refinement steps, as it reflected their internal judgment on the true tumor contour. Thus, the y-axis here can be considered as a progress indicator to reflect when the experts stop refining the contours, not

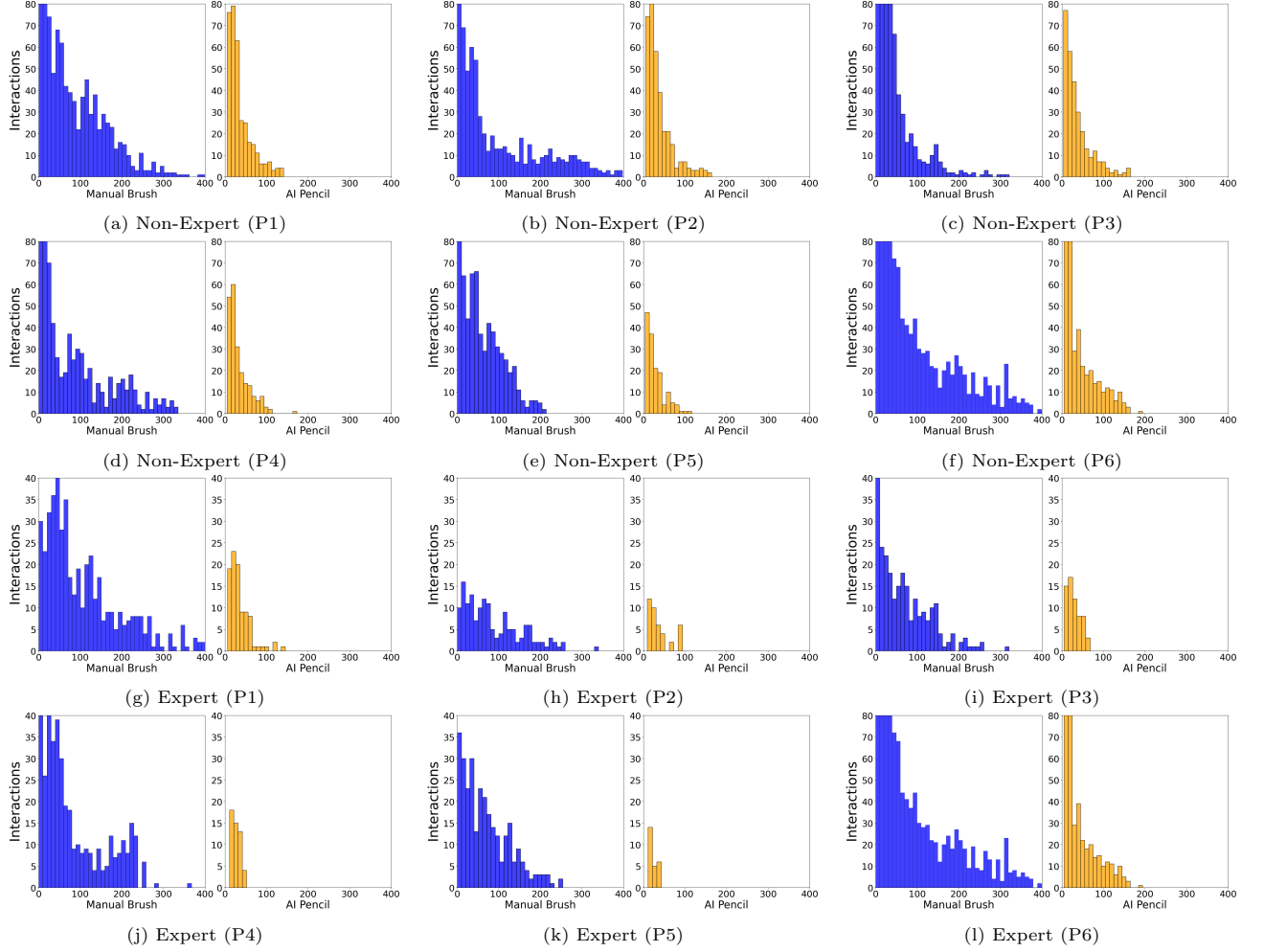


Fig 3: Histogram plots showing pixels drawn per interaction (x-axis) during auto-contour refinement of a patient (P) tumor by non-experts (a-f) and experts (g-l).

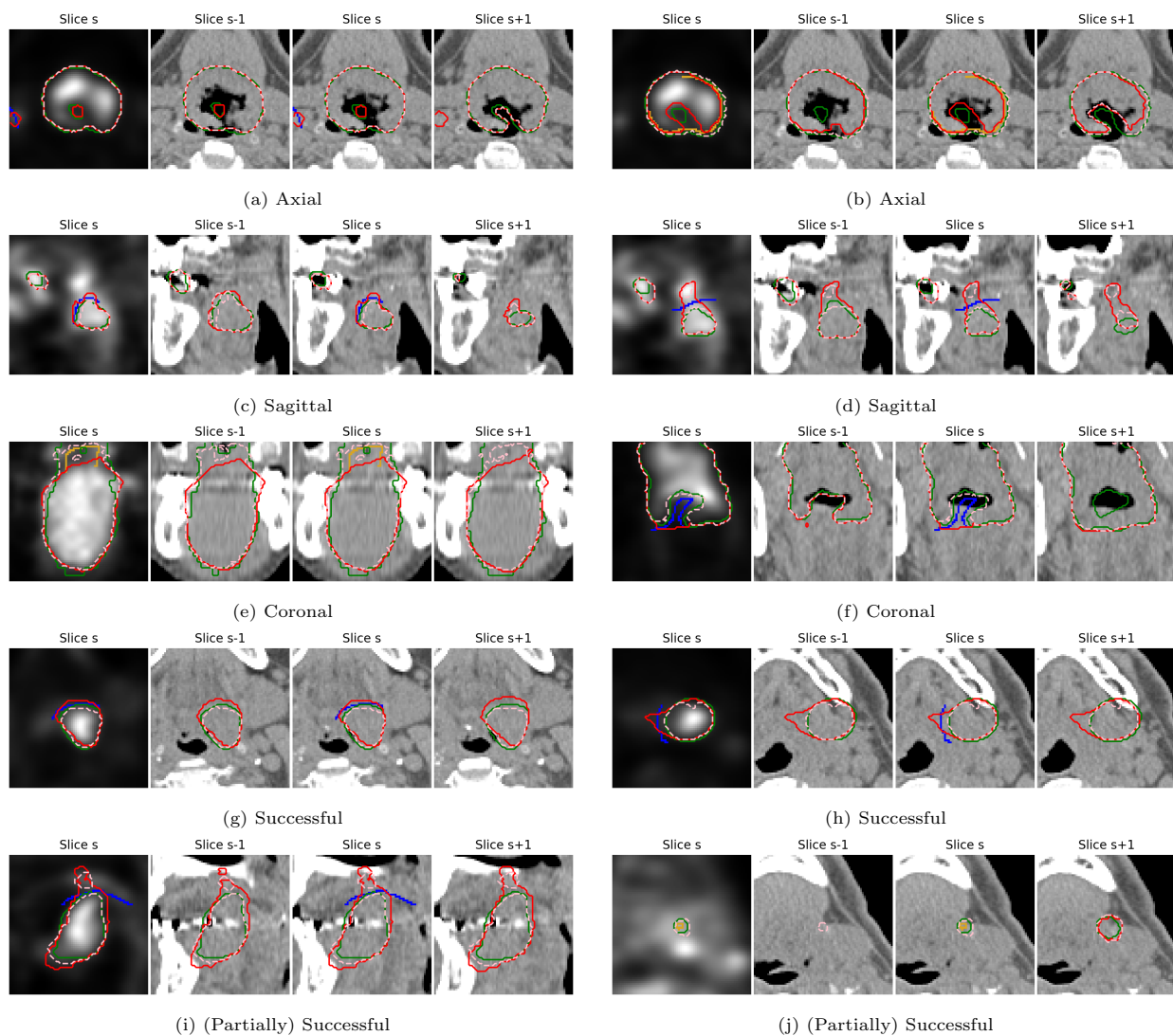


Fig 4: Visual results for AI pencil refinement sessions with PET and CT scans showing previous reference contour (green), predicted contour (red) and refined contour (pink) along with scribble (yellow=tumor, blue=non-tumor). Results are shown for axial (a,b), sagittal (c,d) and coronal (e,f) views. While some scribbles successfully produced a refined contour that matches the reference (g,h), others were only partially successful (i,j).

Table 2: Dice (surface Dice @ 2 mm) when comparing a patient’s (P) final contours across manual brush and AI pencil (a). The same metrics were also used to show inter-observer variability (IoV) for both tools (b). Per-patient scores for each tool (against ground truth) are shown in (c).

(a) Dice (surface Dice @ 2 mm) between the final contours of the tools (for experts).						
	P1	P2	P3	P4	P5	P6
E1	0.78 (0.66)	0.85 (0.74)	0.81 (0.77)	0.67 (0.58)	0.87 (0.92)	0.80 (0.74)
E2	0.85 (0.85)	0.87 (0.82)	0.80 (0.83)	0.71 (0.62)	0.81 (0.80)	0.81 (0.81)
E3	0.87 (0.86)	0.89 (0.85)	0.83 (0.84)	0.74 (0.74)	0.81 (0.85)	0.76 (0.66)
E4	0.84 (0.77)	0.88 (0.83)	0.80 (0.78)	0.78 (0.76)	0.81 (0.74)	0.77 (0.65)
Avg Patient Dice	0.84 (0.79)	0.87 (0.81)	0.81 (0.81)	0.72 (0.68)	0.83 (0.83)	0.79 (0.71)
(b) Dice (surface Dice @ 2 mm) as expert interobserver variability (IoV) across patients.						
	P1	P2	P3	P4	P5	P6
Manual						
Brush	0.81 (0.74)	0.96 (0.91)	0.94 (0.94)	0.89 (0.93)	0.82 (0.85)	0.89 (0.92)
IoV						
AI						
Pencil	0.88 (0.85)	0.95 (0.94)	0.87 (0.89)	0.88 (0.95)	0.87 (0.93)	0.93 (0.95)
IoV						
(c) Average Dice (surface Dice @ 2 mm) between the final expert contour of each tool and dataset ground truth.						
	P1	P2	P3	P4	P5	P6
Manual						
Brush	0.82 (0.75)	0.83 (0.65)	0.88 (0.93)	0.85 (0.84)	0.83 (0.85)	0.85 (0.83)
AI						
Pencil	0.88 (0.88)	0.88 (0.87)	0.83 (0.89)	0.80 (0.79)	0.83 (0.88)	0.80 (0.80)

necessarily as the true quality of the final contour.

To summarize, for the manual brush, it can be seen from all subfigures of Fig. 2 that it slowly but steadily increased contour quality, while the AI pencil increased more sharply and then plateaued. This is because the manual brush could only edit one slice at a time while the AI pencil had the capability of using sparse 2D scribble inputs to refine contours in 3D.

The ItV of the tumor refinement sessions (Table 2a) indicated that the experts produced similar contours for a patient regardless of the tool. These numbers provided a sense of validity to the final contours submitted in this study. In daily clinical practice, experts also expect to receive Magnetic Resonance Imaging (MRI) scans, physician notes, and endoscopy videos. However, since this work used an open dataset (HECKTOR2022 [23]), there was no access to these resources. This could be a potential factor behind the aforementioned ItV. The introduction of additional resources in future studies could reduce it.

The inter-observer variability (Table 2b) indicated the AI pencil led to slightly better standardization of final tumor contours between experts (a difference of 0.01 was ignored here). Thus, the AI pencil could offer both speed and quality for contour refinement. In Table 2c, the average Dice (surface Dice @ 2mm) with the post-hoc reviewed dataset ground truth had a difference of 0.00 (0.04) between the manual brush and AI tools. Note that the Dice coefficient quantifies volumetric overlap between two contours, while surface Dice evaluates how much of the contouring surfaces lie within a specified distance tolerance (e.g., 2 mm). This alleviated concerns that faster refinements lead to poor final contour quality. The contour review revealed that the ground truth of P4 was incorrect (refer Supplementary Material F). However, to avoid a selection bias, this patient was kept as part of the analysis. Its removal led to an average Dice of 0.84 for both tools. This led to the same conclusion that the AI pencil produced contours of a quality similar to the manual brush.

Another important aspect while interpreting contour refinement dynamics, is contextualizing it with the performance of the auto-contouring model. The model in this work achieved a Dice of 0.76 (on the test dataset i.e., HECKTOR2022-France clinic), which was in line with recent multi-clinic validation results in the $[0.74 - 0.80]$ range [32, 33]. Moreover, performance drops to the $[0.6 - 0.7]$ range are common when models are applied out-of-distribution in clinical practice, making this setting representative of real-world workflows. Finally, a simpler UNet architecture ($\sim 1.2\text{M}$ parameters) was chosen instead of more complex models such as nnUNet ($\sim 30\text{M}$ parameters) for both auto-contouring and contour refinement to standardize design choices and ensure fast prediction times across both tasks. Using the same architecture for both tasks also promotes a simpler and more consistent modeling framework. Similarly, the same dataset was used for both auto-contouring and contour refinement, which was advantageous given the often limited availability of annotated medical data. Thus, reusing components (architecture and dataset) reduces the number of design decisions required when deploying auto-contouring systems in clinical settings.

Ideally, auto-contour refinement tools such as AI pencils should be embedded and tested in high-end contouring platforms offered by commercial radiotherapy software. However, none of the widely used commercial software provide capabilities to access their contouring tools via a programmatic interface. Thus, open-source libraries were chosen to create a web-hosted and open-source interaction platform for this contouring study. The AI pencil could also be readily integrated in open-source platforms such as 3DSlicer [34] or Napari [35], however, the web-based framework was instrumental for conducting this study with experts from different institutes.

Previous studies had established proof on the viability of interactive contouring with 2D [9, 14–16], 2.5D [11] and 3D [10, 12, 13, 17–22] models. However, depending on the contouring task, the imaging input could be either 2D (e.g., X-ray, ultrasound, histopathology, fundus scans) or 3D (e.g., CT, MRI, PET). As the field of interactive contouring matures, future work should consider releasing models that are both 2D and 3D capable. Also, progress in computer vision often occurs because of the presence of open benchmark datasets. Previous 2D [16] and 3D [36, 37] datasets consist of unimodal scans (e.g. CT, MRI, PET, fundus, X-ray). However, medical image contouring, especially in radiotherapy, is usually done in a multi-modal manner, such as in this study. Future research will benefit from the curation of multi-modal datasets where interactive tools such as the AI pencil can be tested. To deal with clinical scenarios, where only a CT scan is available, future work should consider training contour-refinement models with- and without the PET scan. Finally, many of the state-of-the-art 3D models capable of iterative contour refinement [12, 13, 19, 22] are large models and could affect inference time. Hence, neural network parameter count is an important factor and should be considered as a factor when running clinical trials on these models.

An important consideration of interactive contouring is that as auto-contouring models improve over time, the use of interactive contouring will reduce. However, there will always be a few auto-contour cases on which clinicians will have to make edits, so such tools will continue to persist. In this work, it was noticed that the AI pencil was best suited for obvious or large false-positive or false-negative regions that could be refined with sparse tumor or non-tumor scribbles. Small isolated errors or ambiguous boundaries may still require the manual brush. Thus, the AI pencil (which would also require short user training) should be viewed as an assistive refinement tool rather than as a complete replacement for manual editing.

Finally, although with 11 users this study is the largest to date, it was conducted on a single demographic (France - CHUP clinic) across 6 patients. Also, the analysis of this study is strongly tied to the use of the HECKTOR2022 dataset as a training cohort for both models. This could introduce potential bias due to any imaging and auto-contouring error/refinement patterns tied to it. Future work should address this limitation of this study and consider expanding the study both in terms of size, scanner variability as well as demographic diversity. Moreover, the use of a self-made auto-contouring model with a lightweight neural architecture can influence results and future work with commercial tumor contouring tools could improve clinical significance.

With a projection for increased occurrence of cancer cases and a shortage of radiotherapy clinicians [38, 39], there is an increasing need for automating the radiotherapy workflow [40]. However, since human supervision is still paramount [41], human-centric AI techniques such as the proposed AI pencil for contour quality refinement will be critical in achieving safety and efficiency standards for high quality radiotherapy care.

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Supplementary Material A. Dataset

The HECKTOR2022 [23] dataset was a grand challenge dataset from the Medical Image Computing and Computer Assisted Intervention (MICCAI) society containing scans from seven different radiotherapy clinics. The clinics within it were abbreviated by the challenge organizers as: Canada (CHGJ, CHUS, CHMR, CHUM), Switzerland (CHUV), United States of America (MDA) and France (CHUP). The auto-contouring model was trained on Canadian (CHGJ, CHUS, CHUM), Swiss (CHUV) and American (MDA) patients. Data from one Canadian clinic was kept aside (i.e., CHMR as in-distribution data) for validation and one French clinic (i.e., CHUP as out-of-distribution data) for testing purposes. CHMR was used for also training users on the interface as well as the use of the AI pencil.

All scans were resampled to an isotropic voxel size of 1 mm using B-spline interpolation, and contours using nearest-neighbor interpolation. For the sake of simplicity, an area of (144,144,144) was cropped around the primary head-and-neck tumor and used that during training and testing of both the auto-contouring and contour-refinement (i.e., AI pencil) models. All scans were normalized using Hounsfield Unit (HU) windowing $[-250,250]$ for the Computed Tomography (CT) scan and standardized uptake value (SUV) windowing $[0,25]$ for the Positron Emission Tomography (PET) scan. Finally, z-normalization of the scans was performed as is the standard practice for AI models.

Supplementary Material B. Auto-contouring and contour-refinement models

Both the auto-contour and contour-refinement models in this work were setup using the Medical Open Network for AI (MONAI) software package [24]. While the auto-contouring model has only a two-channel input (i.e. CT + PET), the contour-refinement model had a five-channel input (i.e., CT + PET + Auto-Contour + Tumor Scribble + Non-Tumor Scribble). The `monai.networks.nets.BasicUNet` class was used. The four internal residual convolutional blocks contained 16, 32, 64 and 128 layers. The models were trained using a standard cross-entropy loss and with the Adam optimizer (fixed learning rate of 0.001).

During training, potential auto-contouring errors were simulated by either deforming the ground-truth contour provided by the HECKTOR2022 dataset or randomly removing chunks from the contour mask. Since the ground truth was available, synthetic scribbles were generated in the false positive or false negative regions to simulate human scribbles. Two types of scribbles were generated: contour-based or morphology-based (via medial axis or skeletonization) similar to previous work [16]. For training data augmentation, a portion of the scribble pixels were sampled and deformations on it were performed to simulate human scribble randomness.

Supplementary Material C. Web interface

For the user interface (Fig. 1a), cornerstone3D [27], a medical imaging library written in the Javascript programming language was used. They provide software components for rendering images and contours in the Digital Imaging and Communications in Medicine (DICOM) standard along with contouring tools such as brushes and pencils. For the scan database, the Orthanc server [29] was used which hosted the DICOM files of CT and PET scans, reference contours and also the auto-contouring predictions. The scribbles (of the AI pencil) provided by the user on the front end were then sent to a back end FastAPI server [28] that used the Python programming language. Here, AI inference was performed to refine the contour based on the AI pencils' scribbles. The refined contour was then sent back to the front end. A video of the interface has been uploaded [30].

Supplementary Material D. Additional results on user effort

The total time taken during an auto-contour refinement session is a combination of the total user interactions, total pixels drawn (Table D.3a, D.3b) and the total slices scrolled (Table D.3c, D.3d). Users also spent time on analysing the output of their previous interaction (either manual brush or AI-pencil), however, these pauses in interactions were not captured.

The tables below show the savings in pixels drawn and slices scrolled with the AI pencil when compared to the manual brush.

Table D.3: Savings in pixels drawn (a,b) and slices scrolled (c,d) when using AI pencil as compared to manual brush. Savings in percentages are shown in brackets.

(a) Pixels drawn savings in non-expert (NE) sessions.

	P1	P2	P3	P4	P5	P6
NE1	11035 (87%)	8925 (77%)	3801 (68%)	9308 (90%)	4885 (85%)	17987 (88%)
NE2	9465 (84%)	9057 (88%)	3566 (75%)	8102 (75%)	4770 (88%)	15753 (90%)
NE3	13501 (90%)	8832 (82%)	3326 (63%)	7568 (88%)	5787 (93%)	13993 (79%)
NE4	11039 (84%)	7918 (77%)	4756 (72%)	8294 (92%)	5941 (91%)	15319 (84%)
NE5	9926 (84%)	7704 (84%)	2803 (68%)	7207 (86%)	4302 (88%)	12542 (81%)
NE6	10803 (88%)	8991 (78%)	3383 (63%)	7044 (89%)	5510 (90%)	15799 (88%)
NE7	8339 (77%)	6802 (75%)	3053 (68%)	6062 (79%)	4199 (84%)	12739 (81%)
Range	[8339,13501] ([84%,90%])	[6802,9057] ([75%,88%])	[2803,4756] ([63%,75%])	[6062,9308] ([75%,92%])	[4199,5941] ([84%,93%])	[12542,17987] ([79%,90%])

(b) Pixels drawn savings in expert (E) sessions.

	P1	P2	P3	P4	P5	P6
E1	9017 (89%)	3230 (81%)	2578 (81%)	9595 (94%)	8252 (98%)	16293 (98%)
E2	16253 (97%)	2981 (92%)	5539 (92%)	7203 (95%)	4569 (95%)	18516 (97%)
E3	14691 (92%)	2595 (89%)	3518 (86%)	8759 (97%)	5991 (97%)	12848 (98%)
E4	10417 (94%)	3406 (94%)	4328 (89%)	7139 (96%)	1103 (96%)	14310 (99%)
Range	[9017,16253] ([89%,97%])	[2595,3406] ([81%,94%])	[2578,5539] ([81%,92%])	[7203,9595] ([94%,97%])	[1103,8252] ([95%,98%])	[12848,18516] ([97%,99%])

(c) Slices scrolled savings in non-expert (NE) sessions.

	P1	P2	P3	P4	P5	P6
NE1	55 (21%)	-239 (-79%)	-130 (-45%)	-78 (-11%)	-520 (-137%)	131 (13%)
NE2	-487 (-123%)	-170 (-33%)	-317 (-85%)	86 (16%)	45 (8%)	-237 (-47%)
NE3	230 (20%)	-127 (-45%)	87 (12%)	-390 (-167%)	32 (7%)	-687 (-470%)
NE4	12 (3%)	-401 (-98%)	-109 (-38%)	-493 (-109%)	-25 (-3%)	-803 (-117%)
NE5	-135 (-23%)	207 (19%)	154 (48%)	-237 (-100%)	-107 (-23%)	-95 (-18%)
NE6	-402 (-230%)	230 (36%)	10 (3%)	-367 (-67%)	52 (17%)	239 (56%)
NE7	-210 (-31%)	-810 (-447%)	391 (37%)	-326 (-51%)	-261 (-46%)	-211 (-26%)
Range	[-487,230] ([-230%,21%])	[-810,230] ([-447%,36%])	[-317,391] ([-85%,37%])	[-493,86] ([-167%,16%])	[-520,52] ([-137%,52%])	[-803,239] ([-470%,56%])

(d) Slices scrolled savings in expert (E) sessions.

	P1	P2	P3	P4	P5	P6
E1	-452 (-178%)	234 (34%)	-323 (-101%)	126 (32%)	266 (72%)	443 (62%)
E2	-37 (-9%)	-91 (-26%)	-56 (-12%)	-328 (-100%)	141 (25%)	300 (39%)
E3	-112 (-20%)	340 (50%)	-190 (-47%)	463 (61%)	95 (24%)	537 (48%)
E4	440 (50%)	251 (38%)	200 (24%)	954 (59%)	246 (64%)	292 (24%)
Range	[-452,440] ([-178%,50%])	[-91,340] ([-26%,50%])	[-323,200] ([-101%,24%])	[-328,954] ([-100%,61%])	[95,266] ([24%,72%])	[292,537] ([24%,62%])

Supplementary Material E. Absolute time taken for experiments

Table E.4: Manual brush and AI pencil annotation times (seconds) for non-experts (U1–U7) and experts (C1–C4) across patients (P1–P6).

(a) Manual brush annotation time (seconds) for non-experts (U1–U7) across patients (P1–P6).						
	P1	P2	P3	P4	P5	P6
U1	684	1570	1128	718	469	1387
U2	480	710	687	440	360	922
U3	558	727	692	416	350	1167
U4	405	530	481	336	263	569
U5	457	649	336	504	332	507
U6	783	632	564	500	488	1346
U7	595	568	569	370	499	708
Range (min, max)	[405, 783]	[530, 1570]	[336, 1128]	[336, 718]	[263, 499]	[507, 1387]
(b) AI pencil annotation time (seconds) for non-experts (U1–U7) across patients (P1–P6).						
	P1	P2	P3	P4	P5	P6
U1	148	552	806	390	373	665
U2	154	307	247	273	248	408
U3	239	325	301	219	127	491
U4	315	405	287	156	133	459
U5	345	325	240	233	131	455
U6	343	138	335	311	148	520
U7	396	447	280	353	253	403
Range (min, max)	[148, 396]	[138, 552]	[240, 806]	[156, 390]	[127, 373]	[403, 665]
(c) Manual brush annotation time (seconds) for experts (C1–C4) across patients (P1–P6).						
	P1	P2	P3	P4	P5	P6
C1	512	259	260	784	463	796
C2	725	436	326	482	424	874
C3	463	206	185	296	319	329
C4	294	152	136	286	66	372
Range (min, max)	[294, 725]	[152, 436]	[136, 326]	[286, 784]	[66, 463]	[329, 874]
(d) AI pencil annotation time (seconds) for experts (C1–C4) across patients (P1–P6).						
	P1	P2	P3	P4	P5	P6
C1	236	124	208	108	55	21
C2	221	129	188	178	63	208
C3	233	55	156	45	81	106
C4	176	81	90	64	11	80
Range (min, max)	[176, 236]	[55, 129]	[90, 208]	[45, 178]	[11, 81]	[21, 208]

Supplementary Material F. Erroneous Ground Truth Contour of Patient 4 (P4)

Patient 4 was retained to avoid post-hoc exclusion of a held-out test case. It is acknowledged that the reference contours were not validated before the study, but rather during the journal review phase, as the peer-reviewed HECKTOR2022 challenge dataset was assumed to be reliable. The erroneous ground truth contour (Fig. F.5a) affected the secondary comparison against ground truth (Table 2c). Since expert users were not shown this contour during refinement, it did not affect the primary metric of this work i.e. total refinement time.

A post-hoc review of all six patients' ground truth contours was done using a contouring tool different than the one used in this study i.e., 3D Slicer [34]. Only the ground truth contours were loaded, so the contour reviewer was not shown the AI contour or previously refined auto-contours. Note that this study (i.e. contour refinements) was first done in late 2024, and the final contour review in Aug 2026 so memorization effects on the patients used in this study could be considered to be minimal. Moreover, during the study, the radiation oncologists were also not shown the ground truth contours. One of the radiation oncologists who participated in this paper (11 years of experience) did the contour review, but a multi-observer review would have been stronger. The review led to a change in the ground truth contour of patient 4 (refer Table F.5). The correction consisted primarily of removing an anatomically distinct portion of the mandible that was included in the original tumor contour. Given the obvious nature of the error, there are unlikely to be any large inter-reviewer variability effects. However, the contour review by a single oncologist could be considered a limitation.

Excluding patient 4 from this ground-truth-only comparison gave average Dice of 0.84 for both the manual brush and AI pencil (Table F.6). Including patient 4 with their rectified contour (Fig. F.5b) also gave an average Dice of 0.84 for both the manual brush and AI pencil (Fig. F.6). Both results supported the same conclusion of comparable final contour quality between the two tools.

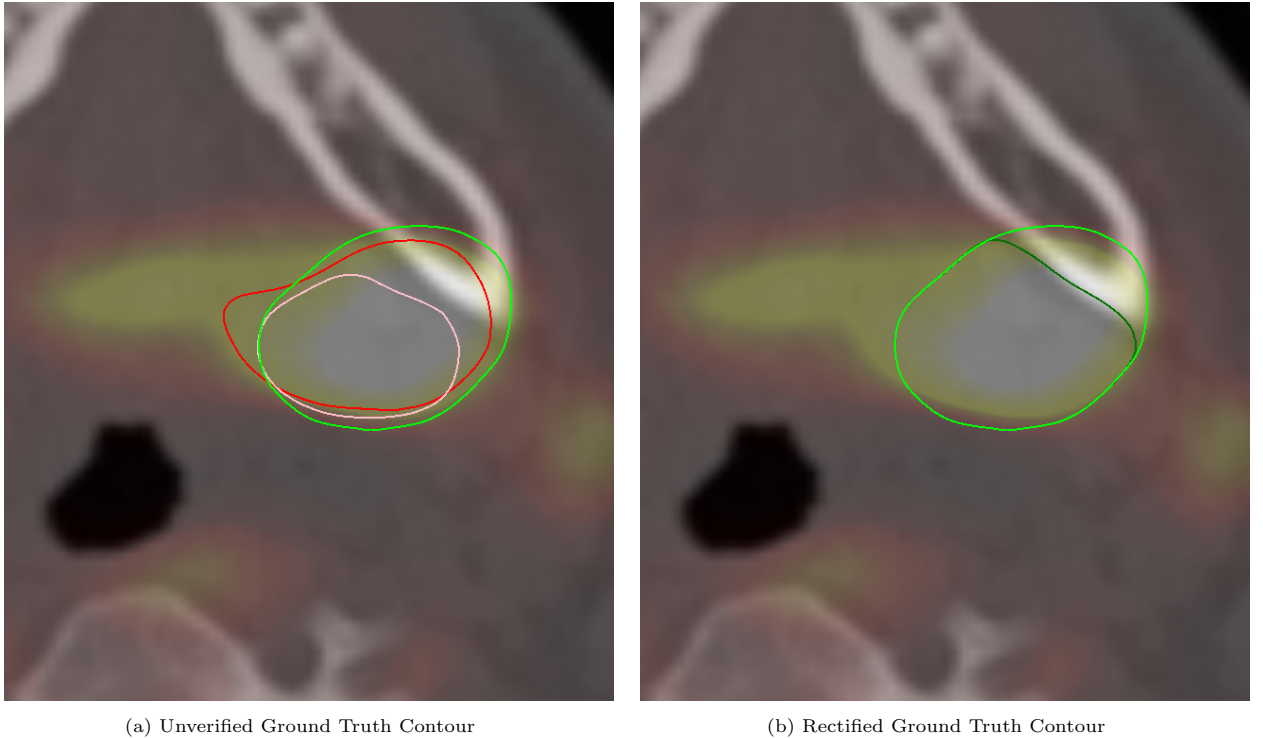


Fig F.5: Erroneous ground truth (light green in (a)), rectified ground truth (dark green in (b)), along with base auto-contour (red) and a refined prediction (pink) for Patient 4 (P4). The image shows a fused CT and PET scan.

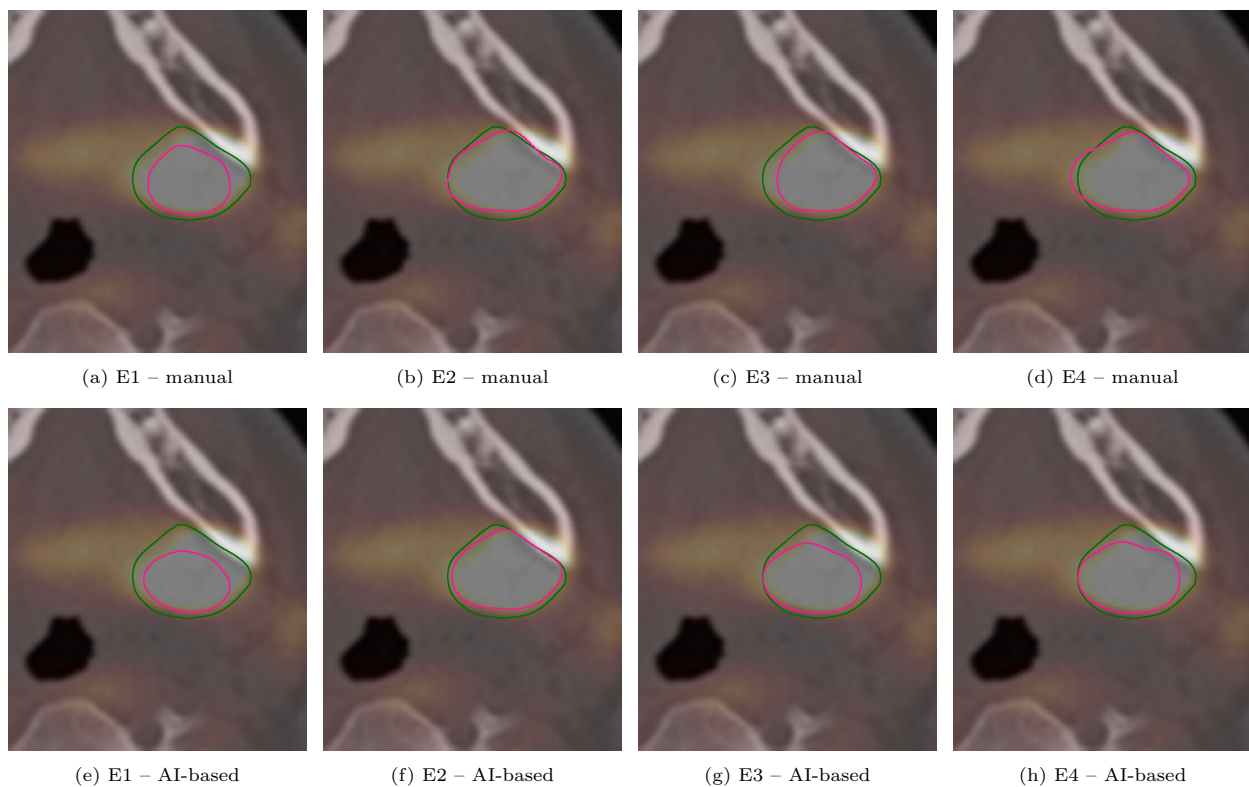


Fig F.6: Rectified ground truth (dark green) and final (refined) tumour contours (in pink) of four experts (E1–E4) for patient P4, shown on a single axial slice (fused CT and PET, weighted 0.8 and 0.2 respectively). The top and bottom rows show the contour obtained after manual brush and AI pencil refinements respectively.

Table F.5: Results of the post-hoc review of ground-truth contours from the HECKTOR2022 dataset. **Within IoV** (inter observer variability) means the contour does not need to be changed.

Patient ID	Verdict
P1	Within IoV
P2	Within IoV
P3	Within IoV
P4	Change needed
P5	Within IoV
P6	Within IoV

Table F.6: Dice (surface Dice @ 2 mm) between the the final expert contour of each tool and unverified dataset ground truth.

	P1	P2	P3	P4	P5	P6
Manual Brush	0.82 (0.75)	0.83 (0.65)	0.88 (0.93)	0.78 (0.78)	0.83 (0.85)	0.85 (0.83)
AI Pencil	0.88 (0.88)	0.88 (0.87)	0.83 (0.89)	0.62 (0.46)	0.83 (0.88)	0.80 (0.80)